Planes and a bird's-eye view

RANSAC. Use a unit-normal plane n.X+d=0. Sample three points, form their 3 x 4 homogeneous matrix, and find the plane as its SVD null space. Reject collinear triples. A point is an inlier if |n.X+d| < 0.025 scene units. Score hypotheses on a fixed random sample of 30,000 remaining points (seed 532). Use an adaptive trial budget targeting 0.999 sampling confidence, with a minimum of 100 and maximum of 2,000 trials. The cap means the confidence target is not guaranteed for small planes. Refine using covariance least squares, classify the full remaining cloud, refine again, and remove that consensus before repeating.

Plane	Full-cloud inliers	Trials	Interpretation
1	2,356,913	102	Background wall
2	854,217	486	Ground
3	407,538	2000	Fountain facade

Ground plane. Plane 2: n=(0.00463422, -0.99871259, 0.05051415), d=1.58323646. Coordinates are in the original camera/world frame, with image y increasing downward. An upright-ground prior (|ny| > 0.85 and positive median y) identifies a candidate; visual inspection confirms the green points cover the floor. Previous planes are absent from later renders. Recolor only current inliers green; do not fill holes.

Camera poses. Image A uses Ra=I, ta=0. Render 768 x 512 with fx=689.875, fy=691.05, cx=380.175, cy=251.70. Project the median ground location onto the plane for the target. Put camera B along the normal above that target at height 8.9328 scene units. Point its optical axis along -n, with x aligned to the projected world-x direction. Choose height from the 98th-percentile ground footprint to leave a framing margin. Use tb=-Rb Cb. The exact K, poses, target, and homography are saved in results/bonus_metrics.json.

Plane-induced homography. Since the first camera is the world frame, H=Kb (Rb - tb n^T/d) Ka^-1 maps A pixels to B pixels. This follows by substituting n.X=-d for points on the ground. Inverse-map destination pixels through H^-1 and implement four-neighbor bilinear interpolation directly. Pixels outside A stay black. No library homography estimation or image warp is used.

Comparison. The ground footprint agrees between directly rendered B and warped A, but the fountain is stretched into the floor by the homography. The 3D renderer retains object height, occlusions, and missing surfaces. The source point cloud was captured from one view, so the overhead render has large holes and sparse surfaces; the warp interpolates existing source pixels. A single plane homography cannot correctly move objects off that plane.

Verification. Tests passed for exact/noisy planes, outliers, degenerate triples, pose orthonormality, sign-invariant homography, and identity, translated, and half-pixel interpolation examples. Projected points placed exactly on the fitted plane agree under H and direct projection to within 3.8e-13 pixels. This is a geometric test, not a claim that the full images match. Run check.py, bonus.py, and bonus_report.py.

Sequential plane detection

1 | Background wall



2 | Ground plane



3 | Fountain facade



Original scene | Image A



Direct rendering versus warping

Image B | Rendered from above



Image A | Ground-plane warp



Image A | Original viewpoint



Warp coverage | White = inside A

